

1(a)(i)	<i>independent:</i> sucrose <u>concentration</u> ; <i>dependent:</i> <u>width</u> of <u>stomata</u> ;	2												
1(a)(ii)	2.5 and 17.5 (to produce a 2.5% sucrose solution) ; 5 and 15 (to produce a 5.0% sucrose solution) ;	2												
1(a)(iii)	1 use all sucrose concentrations ; any four from: 2 leave the epidermis in sucrose solution for the same time ; 3 calibrate, the eyepiece graticule ; 4 <u>measure</u> the width of the stomata ; 5 measure stomata at their widest point ; 6 <i>idea of</i> (measuring) 3 or more (different) stomata (for any single concentration) ; 7 calculate the mean (width of the stomata) for any single concentration ; 8 named hazard <u>and</u> risk <u>and</u> precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>Precaution</th></tr> </thead> <tbody> <tr> <td>plant material</td><td>allergy / irritant</td><td>gloves / eye protection / PPE</td></tr> <tr> <td>slides / cover slip</td><td>cuts</td><td>ensure it is not placed near edge of bench or use glass bin / dustpan and brush / AW, for disposal.</td></tr> <tr> <td>scalpel / knife</td><td>cuts</td><td>cut away from hand / cut onto board / tile / AW</td></tr> </tbody> </table>	hazard	risk	Precaution	plant material	allergy / irritant	gloves / eye protection / PPE	slides / cover slip	cuts	ensure it is not placed near edge of bench or use glass bin / dustpan and brush / AW, for disposal.	scalpel / knife	cuts	cut away from hand / cut onto board / tile / AW	5
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Question	Answer	Marks
1(a)(iv)	<i>predict:</i> (as sucrose concentration increases width of stomata) decreases ; <i>Explain (must match the prediction):</i> (as the sucrose concentration increases) water leaves the <u>guard cells</u>, by, osmosis / down a water potential gradient / from high to low water potential ; or (as the sucrose concentration increases) less water enters the <u>guard cells</u>, by osmosis / down a water potential gradient from high to low water potential ;	2
1(b)(i)	1 stomatal density = $367 \text{ (mm}^{-2}\text{)}$; Any three of: 2 (number of stomata) = 13 ; 3 <u>diameter</u> of picture = $8.5 \text{ cm} / 85 \text{ mm}$ or <u>radius</u> of picture = $4.25 \text{ cm} / 42.5 \text{ mm}$ or <u>length</u> of scale bar / $40 \text{ }\mu\text{m}$ = $1.6 \text{ cm} / 16 \text{ mm}$; 4 show stomatal density calculation ; 5 show magnification (calculation) ;	4
1(b)(ii)	use more than one, field of view / slide / epidermis / (plantain lily) leaf ;	1

Question	Answer		Marks
1(b)(iii)	1 Environmental condition ;	2 Explanation ;	2
	dry / arid / less water	So less water lost / less transpiration	
	low humidity		
	high temperature / warm / hot		
	windy / high wind speed		

Question	Answer	Marks
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